

Curcumin, a component of turmeric: from farm to pharmacy.



Curcumin, an active polyphenol of the golden spice turmeric, is a highly pleiotropic molecule with the potential to modulate the biological activity of a number of signaling molecules. Traditionally, this polyphenol has been used in Asian countries to treat such human ailments as acne, psoriasis, dermatitis, and rash. Recent studies have indicated that curcumin can target newly identified signaling pathways including those associated with microRNA, cancer stem cells, and autophagy. Extensive research from preclinical and clinical studies has delineated the molecular basis for the pharmaceutical uses of this polyphenol against cancer, pulmonary diseases, neurological diseases, liver diseases, metabolic diseases, autoimmune diseases, cardiovascular diseases, and numerous other chronic diseases. Multiple studies have indicated the safety and efficacy of curcumin in numerous animals including rodents, monkeys, horses, rabbits, and cats and have provided a solid basis for evaluating its safety and efficacy in humans. To date, more than 65 human clinical trials of curcumin, which included more than 1000 patients, have been completed, and as many as 35 clinical trials are underway. Curcumin is now used as a supplement in several countries including the United States, India, Japan, Korea, Thailand, China, Turkey, South Africa, Nepal, and Pakistan. In this review, we provide evidence for the pharmaceutical uses of curcumin for various diseases. Copyright © 2013 International Union of Biochemistry and Molecular Biology, Inc.